

Q.28 Determine the focus and directrix of the parabola $y^2 = 12x$. (CO3)

Q.29 Differentiate $y = \frac{(2x-5)}{(x-3)}$ with respect to x . (CO4)

Q.30 Find the second-order derivative $\frac{d^2y}{dx^2}$ for $y = x^3 + 6x^2 - 7x + \cos x$. (CO6)

Q.31 Evaluate $\int x \cos x \, dx$ using integration by parts. (CO5)

Q.32 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ (CO4)

Q.33 Solve the differential equation $\frac{dy}{dx} = \frac{y}{x}$. (CO6)

Q.34 Find the area bounded by the curve $y = x^2 + 1$ and x -axis between the lines $x = 1$ and $x = 3$. (CO5)

Q.35 Evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$. (CO5)

Section-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Find the points of maxima and minima and their corresponding maximum and minimum value of the function $f(x) = x^3 - 3x^2 + 3x - 2$. (CO4)

Q.37 Use Simpson rule to evaluate $\int_1^7 (2x+3) \, dx$ by taking 6 equal sub-intervals. (CO5)

Q.38 Prove that $\sin 10^\circ \sin 50^\circ \sin 60^\circ \sin 70^\circ = \frac{\sqrt{3}}{16}$. (CO3)

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Section-A

Note: Multiple Choice questions. All questions are compulsory. (10x1 = 10)

Q.1 The n^{th} term of an Arithmetic Progression (AP) with first term a and difference d is (CO1)

(a) $a + (n+1)d$ (b) $a + (n-1)d$

(c) ar^{n-1} (d) $\frac{n}{2}[2a + (n-1)d]$

Q.2 In the binomial expansion of $(x+a)^n$, the number of term is: (CO1)

(a) n (b) $(n-1)$

(c) $(n+1)$ (d) $\frac{n}{2}[2 + (n-1)]$

Q.3 The value of $\sin 45^\circ$ is: (CO2)

(a) 1 (b) $\frac{1}{2}$

(c) $\frac{1}{\sqrt{2}}$ (d) 0

Q.4 The angle 188° in which quadrant? (CO2)

(a) 1st (b) 2nd

(c) 3rd (d) 4th

Q.5 The distance between points (0,0) and (3,4) is: (CO3)

- (a) 5 (b) 3
(c) 4 (d) 7

Q.6 The derivative of e^x with respect to x is: (CO4)

- (a) xe^{x-1} (b) $\log x$
(c) e^x (d) $\frac{1}{x}$

Q.7 Evaluation of $\int \cos x \, dx$ results in: (CO5)

- (a) $\sin x + C$ (b) $-\sin x + C$
(c) $\tan x + C$ (d) $\sec x + C$

Q.8 $\int \frac{1}{x} \, dx =$ _____ (CO5)

- (a) $\sin x + C$ (b) $\log x + C$
(c) 0 (d) none of these

Q.9 $225^\circ =$ _____ Radian (CO2)

- (a) $\frac{3\pi}{4}$ (b) $\frac{5\pi}{4}$
(c) $\frac{7\pi}{4}$ (d) $\frac{9\pi}{4}$

Q.10 Radius of the circle $(x-2)^2 + y^2 = 9$ is (CO3)

- (a) 2 (b) 9
(c) 0 (d) 3

Section-B

Note: Objective/Completion type questions. All questions are compulsory. (10x1 = 10)

Q.11 Find the 10th term of the sequence 2, 4, 6, 8, _____. (CO1)

Q.12 Convert 90° in to radians. (CO2)

Q.13 Write the general equation of a circle. (CO3)

Q.14 State the formula for the Centroid of a triangle. (CO3)

Q.15 Evaluate the limit : $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ (CO4)

Q.16 What is the derivative of $\log x$? (CO4)

Q.17 Order of differential equation $\frac{dy}{dx} + y = 0$ is _____. (CO6)

Q.18 Define a Homogeneous Differential Equation. (CO6)

Q.19 $\frac{d^2y}{dx^2} + 5y = 0$ is a linear differential equation. (True/False) (CO6)

Q.20 Midpoint of (3,4) and (5,6) is _____. (CO3)

Section-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5 = 60)

Q.21 Resolve $\frac{1}{(x+1)(x+2)}$ into partial fractions. (CO1)

Q.22 Expand $(x-1)^5$ using binomial theorem. (CO1)

Q.23 Find the value of $\tan 105^\circ$. (CO2)

Q.24 Prove that $\frac{\sin x}{1 + \cos x} = \tan \frac{x}{2}$ (CO2)

Q.25 Find the equation of the circle that has center (2,3) and passes through (5,7). (CO3)

Q.26 Find the slope and equation of the line which passes through the points (2,-6) and (-5,1). (CO3)

Q.27 Find the radius and centre of given equation of the circle.
 $x^2 + y^2 + 4x + 6y - 3 = 0$ (CO3)